

# DYNAMIC COLLEGE ADMISSIONS

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We study the relevance of incorporating dynamic incentives and eliciting private information about students' preferences to improve their welfare and downstream outcomes in centralized assignment mechanisms. Using administrative data and two nationwide surveys, we identify two behavioral channels that largely explain students' dynamic decisions: (i) initial mismatches and (ii) learning. Based on these facts, we build and estimate a structural model of students' college progression in the presence of a centralized admission system, allowing students to learn about their preferences and abilities over time and reapply to the system. We use the estimated model to analyze the impact of changing the information environment and the assignment mechanism and reapplication rules on the efficiency of the system. Our counterfactual results show that policies that front-load learning can help to reduce switches, while providing score bonuses that elicit information on students' cardinal preferences and leverage dynamic incentives can significantly increase retention and students' overall welfare.

KEYWORDS: college admissions, dynamic matching, college retention.

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## 1. INTRODUCTION

According to [Kapor et al. \(2024\)](#), at least 46 countries—including Turkey, Taiwan, Tunisia, Hungary, and Chile—use centralized systems to organize college admissions. Although extensive literature analyzes the pros and cons of different mechanisms used to perform the allocation, their effects on downstream policy-relevant outcomes is unclear. For instance, only 40% of full-time bachelor’s students graduate on time in developed countries (OECD 2019), and in Chile, just 16% do so, with 30% switching programs and another 30% dropping out. These patterns illustrate potential inefficiencies in the assignment process, suggesting that better-designed assignment mechanisms and information environments could substantially improve downstream educational outcomes.

Improving outcomes requires accounting for features that characterize real-life applications and that are generally overlooked in the literature. One such feature is that matching markets are typically dynamic. For instance, students can learn over time about their preferences and abilities, and this may influence their decisions to reapply, switch to another program, or drop out. Another feature is that students may have private information that is not elicited in the admissions process and could affect their future outcomes and the higher education system’s efficiency. For instance, students’ intrinsic motivation or vocation, which would be captured by their cardinal (or the intensity of their) preferences, could affect their persistence in their programs and, thus, impact the system’s yield. Therefore, designing admissions systems that consider the dynamic interplay between learning, decisions and elicit information about students’ cardinal preferences can be critical for improving students’ outcomes and the efficiency of the system.

In this paper, we study the design of matching markets where agents face dynamic considerations, learn about their abilities and preferences through experience, hold private information that affects outcomes, and may hold biased beliefs about their abilities and future earnings. We examine how changes to the information environment and the assignment mechanism affect students’ welfare and downstream outcomes—including college performance, retention, and on-time graduation—by facilitating learning and incorporating dynamic incentives that leverage students’ cardinal preferences.

By combining data from the Chilean college admissions system and multiple nationwide surveys that we designed and conducted, we show that two behavioral channels largely explain students' dynamic decisions. The first, called the *learning* channel, posits that students learn about their preferences and abilities during their college experience, affecting both their consumption value and expected returns, and potentially motivating them to switch programs or drop out to avoid ex post mismatches. The second, called the *initial mismatch* channel, suggests that students may initially enroll in less-preferred programs to improve their future outside options and later switch to more desirable ones.

These two channels may have different implications. On the one hand, if learning is limited and thus preferences are persistent over time, it may be desirable to discourage reapplications by forcing students to internalize the crowd-out externality they generate—that is, when a student is admitted to an over-demanded program, they may displace another student who would have otherwise secured that spot and is instead assigned to a less preferred option. If the first student later decides to switch programs, the seat they leave behind often goes unfilled, creating an externality that negatively affects others in the system. On the other hand, if learning explains most students' dynamic decisions, it may improve welfare to facilitate switching and avoid ex post mismatches. Hence, the welfare implications of limiting or encouraging switching are unclear. We illustrate this tension through a stylized example in Appendix A, which shows how both channels affect students' decisions and welfare, and how inefficiencies can be mitigated by eliciting cardinal preferences.

To evaluate the effects of these two channels, we introduce a structural model that captures the application behavior of students, as well as their decisions to enroll, retake the admission tests, reapply, switch, and drop out, allowing students to learn about their unknown abilities and their preferences during their academic progression. In particular, we assume that students base their application and enrollment decisions on both (i) the value of studying each program, (ii) the continuation value of retaking the admission tests and reapplying to the system, and (iii) their labor market prospects. Students have heterogeneous preferences—which evolve to capture persistent idiosyncratic shocks that occur each period—and potentially biased beliefs over their future college grades and labor market

prospects—which they update as they observe noisy signals of their unknown ability from their grades. Based on their updated continuation values, students decide whether to continue in their current program, reapply to the system, or drop out and choose their outside option. Finally, students face graduation probabilities, and then enter the labor force and receive pecuniary and non-pecuniary values from the labor market.

The main challenge in estimating our model is separately identifying the *learning*—which includes learning about their abilities from grades and their preferences about their major of enrollment—and the *mismatch* channels. To identify the former, we leverage the longitudinal structure of the data, focusing on how students update their beliefs and change their preferences over time. We isolate learning about preferences by examining the composition of initial applications and reapplications following non-informative grade signals, where any preference change likely reflects evolving tastes rather than updated beliefs about ability. In contrast, we identify learning about unknown abilities by using elicited beliefs about academic performance (e.g., expected grades) and their correlation with (i) changes in the composition of preferences and (ii) decisions (e.g., switching or dropping out) after receiving first-year grades. The correlation between these belief updates and switching behavior and their heterogeneity across majors and subjects help us distinguish between belief-based learning and preference shifts. To identify the initial mismatch channel, we exploit quasi-experimental variation around admission cutoffs, which allows us to compare students who are initially assigned to different programs despite similar application behavior and observed characteristics. These initial differences in assignment—combined with subsequent outcomes such as switching, dropout, or reapplication—provide clean variation to identify the consequences of mismatch separately from endogenous learning dynamics. For instance, we show that there is a strong causal effect of not being assigned to the top-reported preference on the probabilities of reapplying and switching (63% and 51% increase, resp.), supporting the existence of the mismatch channel.

After estimating the structural model, we conduct a series of counterfactual analyses. First, we find that learning accounts for about one-third of all switching decisions, while initial mismatches and congestion explain the remainder and part of dropout decisions.

1 Next, we examine how changes in (i) the information environment and (ii) the assign- 1  
 2 ment process—through changes in the reapplication rules or in the mechanism—affect stu- 2  
 3 dents’ outcomes. On the information side, we find that enabling students to front-load their 3  
 4 learning process (through counseling during high-school or a mandatory one-year post- 4  
 5 secondary program) can help reducing switching rates. On the mechanism side, we find 5  
 6 that penalizing students who switch (as in Turkey); giving a score bonus for all first-year 6  
 7 applicants (as in Finland); or allowing students to signal one of their preferences to get a 7  
 8 score bonus (as in some job markets) can significantly reduce switching rates and increase 8  
 9 students’ welfare. Finally, we find that these effects are robust to changes in the share of 9  
 10 strategic participants, unlike policies such as restricting application list lengths, which are 10  
 11 widespread but far less effective and sometimes detrimental for students’ outcomes. 11

12 Our counterfactual experiments highlight the importance of carefully balancing the two 12  
 13 key behavioral forces at play: the benefits of learning through experimentation and the 13  
 14 crowd-out externality created by initial mismatches. We show that *facilitating learning*, 14  
 15 incorporating dynamic incentives, and eliciting students’ cardinal preferences—through 15  
 16 modifications to reapplication rules and the assignment mechanism—can meaningfully in- 16  
 17 fluence both educational outcomes and overall welfare. More broadly, these findings un- 17  
 18 derscore the relevance of accounting for dynamic behavior and externalities in the design 18  
 19 of matching markets, with potential implications for improving efficiency and equity in 19  
 20 contexts such as school choice, organ allocation, and entry-level labor markets. 20

21 The remainder of the paper is organized as follows. Section 2 discusses the most closely 21  
 22 related literature. Section 3 describes the Chilean college admissions system and provides 22  
 23 empirical evidence for the two behavioral channels. Section 4 presents our model. Section 5 23  
 24 describes our identification strategy. Section 6 describes the estimation approach and its 24  
 25 results. Section 7 reports our counterfactual results, and Section 8 concludes. 25

## 27 2. LITERATURE 27

28  
 29 Our paper combines two strands of the literature: (i) the empirical analysis of assignment 29  
 30 mechanisms and (ii) the empirical analysis of college choices under uncertainty. 30

The first strand focuses on understanding the incentives that centralized assignment mechanisms introduce; how to use the data generated by these mechanisms to identify and estimate students' preferences/beliefs (Agarwal and Somaini, 2019); and measuring the welfare effects of changing assignment mechanisms in different settings (Agarwal and Somaini, 2018, Calsamiglia et al., 2020, He, 2012, Kapor et al., 2020, Luflade, 2017, Hernandez-Chanto, 2021).

Despite progress in this area, the aforementioned studies either consider static settings, assume fixed preferences over time, or simply ignore the effects of assignment mechanisms on downstream outcomes.<sup>1</sup> In response, a recent line of research has begun to examine dynamic mechanisms or settings in which agents interact repeatedly with the assignment process, including the allocation of public housing (Waldinger, 2021), hunting licenses (Reeling and Verdier, 2021), general practitioners (Huitfeldt et al., 2025), teachers' labor markets (Ederer, 2023), and kidneys from deceased donors (Agarwal et al., 2024a,b). In educational markets, dynamic models have been recently used to evaluate the impact of middle school placements on high school outcomes (Hahm and Park, 2022), day-care assignments (Kuno, 2023, De Groote and Rho, 2024), and sequential college admissions (De Groote et al., 2025). Among these, the closest to ours is (Narita, 2018), who analyzes theoretically and empirically the welfare effects of centralized school-choice mechanisms with rematching when demand evolves over time. We contribute to this literature by developing a dynamic framework for centralized college admissions that captures how different forms of learning, strategic behavior, and repeated interactions with the mechanism shape system yield, student outcomes, and welfare.

The second strand studies education and occupation choices, focusing on how students gather information and learn about their preferences, abilities and future earnings (Arcidiacono, 2005, Arcidiacono et al., 2025, Malamud, 2011) and how their choices impact outcomes and returns (Altonji et al., 2012, 2016). Within this strand, the closest paper to ours is Bordon and Fu (2015), who analyze the effects of switching from a system where stu-

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<sup>1</sup>With some exceptions in school reforms (Tanaka et al., 2020), college admissions (Otero et al., 2021), kidney allocation (Agarwal et al., 2024a), and the medical match (Friedrich et al., 2024).

dents choose both college and major simultaneously to one where they choose college first and then major. Our paper complements their work by studying the effects of changing the assignment mechanism and reapplication rules in a dynamic context, allowing for heterogeneous and potentially biased subjective beliefs—identified with survey data—rather than imposing rational expectations, identifying their effect on students’ learning about their abilities and their preferences, and measuring its effect on student outcomes.

Finally, within these two strands of the literature, we also contribute to recent papers exploiting survey data for identification of structural models in college admissions (Wiswall and Zafar, 2015a, 2021, Tincani et al., 2023) and school-choice (Kapor et al., 2020). We add to this literature by leveraging students’ subjective and potentially biased beliefs over their college outcomes and exploiting this information to identify their learning process.

To the best of our knowledge, this is the first paper to structurally evaluate how the information environment and market design choices affect students’ welfare and outcomes—such as achievement, retention, and on-time graduation—beyond the initial match. We also contribute by revisiting the trade-off between eliciting preference intensity and ensuring *strategy-proofness* in a dynamic framework that incorporates private information, learning about abilities and preferences, and heterogeneous, subjective, and potentially biased beliefs about ability and future outcomes. Accounting for these elements is essential to understand how changes in the assignment mechanism influence students’ indirect utilities and welfare, and to evaluate intrinsically dynamic policies—such as reapplication rules—designed to improve both student outcomes and the system’s yield.

### 3. COLLEGE ADMISSIONS IN CHILE

The college admissions process in Chile is semicentralized, with the most selective universities having a centralized system and the remaining institutions conducting their admissions processes independently. This paper’s empirical application follows the cohort of 2014 and focuses on the centralized part of the system, known as *Sistema Único de Admisión* (SUA). This part of the system is organized by the *Consejo de Rectores de las Universidades Chilenas* (CRUCH), and its admissions process is operated by the *Departamento de Evaluación, Medición y Registro Educacional* (DEMRE).

To apply to any of the close to 1,500 academic programs offered by the 41 universities that are part of the centralized system, students must undergo a series of standardized tests (*Prueba de Selección Universitaria* or PSU). These tests include Math, Language, and a choice between Science or History, and provide a score for each. Students' performance during high school yields two additional scores: one obtained from the average grade during high school (NEM) and a second that depends on the relative position of the student among their cohort (Rank). Admission to any program is solely based on these *admission factors*.

After scores are published, students can submit—at no cost—a list with no more than ten programs, ranked in strict order of preference. We refer to these lists as Rank Order Lists (ROLs). Notice that students directly apply to a program, i.e., they must list university-major pairs on their ROL. In the remainder of the paper, we refer to these pairs as programs.

On the other side of the market, each program announces its vacancies, the weights on each admission factor, and the set of additional requirements they will consider for applications to be valid. For instance, universities may require a minimum application score or a minimum score in some PSU tests, among other less common requirements. Each program's preference list is defined by first filtering all applicants who do not meet these requirements. Students are then ordered based on their application scores, which are computed as the weighted sum of the applicants' scores and the predefined weights.

Considering the vacancies and the preferences of the applicants and programs, DEMRE runs an algorithm to match students to programs. The mechanism used is a variant of the student-proposing Deferred Acceptance algorithm, in which all students tied for the last seat in a program are admitted (Rios et al., 2021). A description of the mechanism can be found in Appendix B.1. As a result of the assignment process, each program is associated with a cutoff, such that all students whose weighted score is above it are granted admission, and all students with scores below the cutoff are waitlisted and thus may have to enroll in a lower-ranked preference. This property is known as the *cutoff structure*.

Enrollment begins with the released of assignment results and unfolds in two rounds. In the first, only assigned students may enroll in their designated program. In the second,



1 programs with remaining seats invite waitlisted students to enroll. Throughout the process, 1  
2 applicants may also choose to enroll in non-centralized programs or enter the workforce. 2

3 Students may retake the admissions test and reapply to the centralized system in sub- 3  
4 sequent years. However, retaking the test is not required for those who last took it in the 4  
5 previous year. For students who take the test in two consecutive years, the system allows 5  
6 them to choose the better of the two score pools—i.e., they cannot combine individual test 6  
7 scores across years—for each program to which they apply. Although some students switch 7  
8 programs without formally reapplying, the vast majority of switches—approximately 8  
9 70%—occur through reapplication. Therefore, in the remainder of the paper, we focus on 9  
10 switches that follow a reapplication and abstract from internal transfers, capturing exactly 10  
11 how the centralized system works. Finally, seats that students vacate when they switch are 11  
12 not reallocated to other students nor included in the offer of the next year, so there may be 12  
13 programs that are under-utilized ex-post despite of being oversubscribed ex-ante. 13

### 14 3.1. *Data* 14

15  
16 Our dataset includes (i) administrative data provided by DEMRE and the Ministry of 16  
17 Education (MINEDUC), (ii) several surveys designed and conducted in collaboration with 17  
18 CRUCH and DEMRE aiming to elicit students' preferences and beliefs about admissions 18  
19 probabilities, and (iii) grade records facilitated by CRUCH. Specifically: 19

- 20 • Admissions process: This includes students' socioeconomic characteristics—including 20  
21 self-reported family income, parents' education, and the municipality in which the stu- 21  
22 dent lives, among others—; scores; applications; final assignment; and enrollment de- 22  
23 cisions, spanning from 2007 to 2023. In addition, this includes data on programs char- 23  
24 acteristics, including their vacancies, weights, tuition, duration, major, location, etc. 24
- 25 • Labor market: This includes aggregate information about the labor market outcomes 25  
26 in each program—including average wages four years after graduation and overall em- 26  
27 ployment probabilities one year after graduation—spanning from 2014 to 2018. We 27  
28 also have data at the major level, including average wage for the first five years after 28  
29 graduation; employment probabilities for the first and second year after graduation; 29  
30 and the evolution of average wages from the first ten years after graduation. 30

1 • Grades: This includes the cumulative GPA for their first three years of college for every 1  
2 student who enrolls in a program that is part of the centralized system in 2014 and 2015. 2  
3 • Surveys post-application: This includes responses to surveys that we designed and con- 3  
4 ducted in 2019 and 2020 to gather information on students' preferences—specifically, 4  
5 about their top-true preference—and their beliefs about their admissions probabilities 5  
6 for each program on their ROL and the elicited top-true preference. As many students 6  
7 reapply to the centralized system after a year, these two surveys provide a panel for 7  
8 reapplicants' preferences and their beliefs (see Appendix B.2 for more details). 8  
9 • Surveys pre and post-application: To complement the previous surveys, in 2023, we 9  
10 designed and conducted similar surveys before and after the application time window. 10  
11 In the latter, we ask students about their beliefs (about admission chances, grades, 11  
12 wages, among others) for programs both included in their ROL and also outside it, 12  
13 allowing us to capture beliefs on choices and also on counterfactual programs. 13  
14 To our knowledge, this is the first paper to use the grades data described above and to 14  
15 incorporate nationwide-survey data on students' true preferences, beliefs about admission 15  
16 probabilities, college progression, and future labor-market outcomes. 16  
17 Throughout the paper, we focus on a subset of the population to reduce computational 17  
18 complexity. Specifically, we focus on students who graduated from a high school within 18  
19 the Region Metropolitana (RM) in 2013, participated in the 2014 admissions process (i.e., 19  
20 took the PSU tests), had an average score between Math and Language above 475, and 20  
21 applied to programs located in RM. The latter allows us to reduce the number of programs 21  
22 to less than half (435) without major loss, as close to 80% of applications from students 22  
23 living in RM include only programs located in that region. Finally, we exclude students 23  
24 with average score below 475 (less than 13% of students who can apply) because they 24  
25 do not satisfy loan eligibility requirements. Contrary to previous evidence (Lochner and 25  
26 Monge-Naranjo, 2012, 2016, Solis, 2017), Card and Solis (2020) show that enrollment 26  
27 decisions and college completion rates among students whose average score exceeds 475 27  
28 (thus, eligible for scholarships) are not highly sensitive to differences in prices. Hence, we 28  
29 can ignore potential effects of credit constraints on students' dynamic choices. 29  
30 30

### 3.2. Empirical Facts

We posit that students' dynamic decisions are largely explained by two behavioral channels: (i) *initial-mismatching*, whereby students assigned to less preferred programs reapply to improve their allocation, and (ii) *learning*, whereby students learn about their match-quality and preferences over time and potentially decide to move to other programs. In this section, we provide empirical evidence that supports the existence of these two channels.

*Mismatching.* In Table I, we report the average switching and dropout rates in the first four years, separating by income level—high or low—and gender.<sup>2</sup> First, we observe that close to 21.2% of students switch from their first program of enrollment, and 15.3% drop out within the first four years. Second, comparing outcomes by gender (within an income level), we observe that women are more persistent in their academic progression, since their switching and dropout rates are lower than those for men. On the other hand, comparing these rates by income level (within gender), we observe that low-income students are less likely to switch programs during their academic progression. However, we also observe that low-income students are significantly more likely to drop out. These patterns—suggesting large differences in switching and dropout rates by gender and income—are similar to those observed for first-year outcomes. Therefore, for simplicity, we restrict our analysis to switching and dropout within the first year. This choice is without major loss of generality, since close to 80% of switching takes place in the first two years, and roughly 2/3 of those happen in the first year (see Figure B.4 in Appendix B.4).

To evaluate the impact of assignment rank on student outcomes, Figure 1 presents first-year switching and dropout rates by preference of assignment. We observe that students assigned to lower reported preferences switch at higher rates compared with students assigned to their top reported preference. For instance, among students assigned to their top reported preference, 9.86% switch programs at the end of their first year, compared with

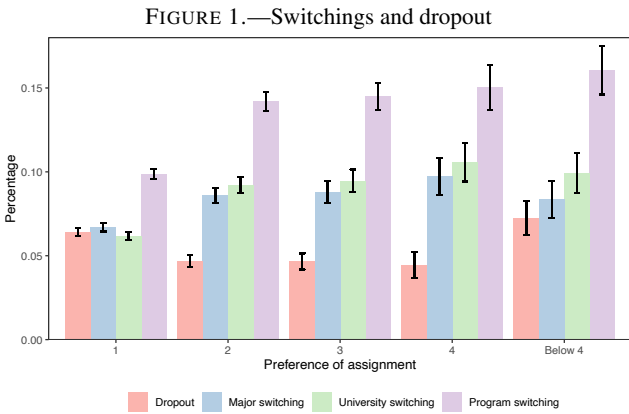
<sup>2</sup>We refer to *majors* as the fields of education provided by the International Standard Classification of Education (ISCED) (UNESCO (2012)) and adapted for Chile, which classifies programs into: Farming, Art and Architecture, Science, Social Sciences, Law, Humanities, Education, Technology, Health, Management, and Commerce.

TABLE I  
SWITCHING AND DROPOUT BY GENDER AND INCOME

|         |        | Switches |            |         |           |         |
|---------|--------|----------|------------|---------|-----------|---------|
|         | Income | Program  | University | Major   | Math type | Dropout |
| Men     | Low    | 0.213    | 0.108      | 0.099   | 0.038     | 0.232   |
|         |        | (0.007)  | (0.005)    | (0.005) | (0.003)   | (0.007) |
|         | High   | 0.24     | 0.128      | 0.132   | 0.044     | 0.134   |
|         |        | (0.006)  | (0.004)    | (0.005) | (0.003)   | (0.005) |
| Women   | Low    | 0.173    | 0.089      | 0.095   | 0.042     | 0.186   |
|         |        | (0.006)  | (0.005)    | (0.005) | (0.003)   | (0.007) |
|         | High   | 0.209    | 0.102      | 0.125   | 0.063     | 0.096   |
|         |        | (0.006)  | (0.004)    | (0.005) | (0.003)   | (0.004) |
| Overall |        | 0.212    | 0.109      | 0.116   | 0.048     | 0.153   |
|         |        | (0.003)  | (0.002)    | (0.002) | (0.002)   | (0.003) |

Note: Standard errors reported in parenthesis. Program denotes any change in the university–major pair; University, Major, and Math-type switches are not mutually exclusive.

almost 15% of those assigned to their fourth choice. These results suggest a strong correlation between assignment rank and switching behavior.

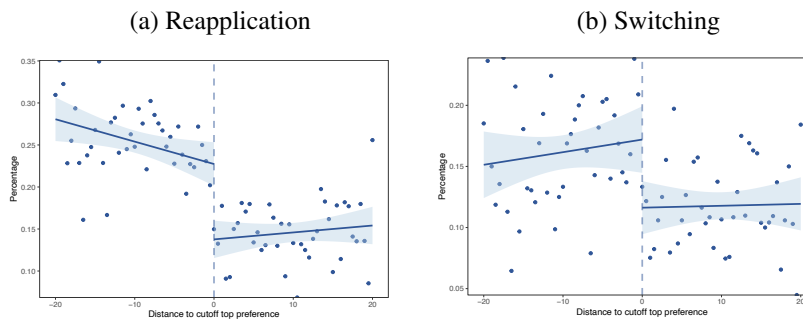


Note: Switching categories do not include stop out.

To make a causal claim, we use a regression discontinuity design that exploits the algorithm’s cutoff structure to perform the allocation. If we assume that students around the cutoff are similar and only differ in their right to enroll in a higher preference, we can estimate the causal effect of interest. In Figure 2, we display binned means of different

outcomes as a function of the distance between cutoffs and students' scores in their most preferred listed program. Figure 2a shows that students right below the cutoff are 9% more likely to reapply the following year, which corresponds to a relative change of close to 62.8% compared to those right above the cutoff. Figure 2b shows that students below the cutoff are 5.8% more likely to switch programs within the centralized system, which corresponds to a relative change of more than 50.9%. Similar results hold when we consider students' top true preferences instead of their top-reported preferences, as shown in Appendix B.3.1. These results confirm our previous findings, i.e., that students assigned to lower preferences are more likely to reapply and switch programs the following year. In Appendix B.3, we provide more details and additional evidence of the effect of crossing the cutoff of the top-reported preference on students' decisions. Importantly, we find no causal effect on enrollment, and, contrary to what Figure 1 suggests, we find no causal effect of cutoff crossing on dropout.

FIGURE 2.—Effect of Cutoff Crossing



The previous empirical facts show a causal effect of the preference of assignment on students' persistence in their initial assignment. To show that the mismatch channel partially explains this, we use the 2020 survey on students' preferences and beliefs, in which we find that a significant fraction of students know, before enrolling in their assigned programs, that they will be less likely to remain in that program if they are assigned to a lower reported preference (see the details in Appendix B.4.1). These results cannot solely be explained by

students' or programs' characteristics and, consequently, we consider them as evidence of match-effects (on students beliefs) about their future college progression.

*Learning.* Students' preferences and beliefs may change during their first year in college, which could affect their decision to switch. We analyze students' switching decisions and classify them into three categories: (i) *Up*, (ii) *Down*, and (iii) *Out*. Students move *Up* (*Down*) if they switch to a program listed above (below) their initial enrollment on their initial ROL. Students move *Out* if they switch to a program not listed on their initial ROL.

We find that, among students who switch in their first year, 17.3% move *Down*, 14.2% move *Up*, and 67.1% move *Out*. The remaining 1.4% of switchers were initially enrolled in programs outside their original ROL (through special admission channels) and subsequently switched. Moreover, more than half of the latter switches involve more selective programs, i.e., programs with higher cutoffs compared to their initial enrollment. These results suggest that both channels explain students' switching significantly. Students who move *Down* or *Out* to less selective programs are likely to have learned about their (poor) ability (learning channel), and students' who move *Up* or *Out* to more selective programs may be trying to find a better match given their initial preferences.

In Table II, we analyze the effect of the signal received at the end of the first year—i.e., the difference between the obtained GPA and the expected one—on reapplication and switching decisions. In all of these models, we control for demographics (gender, income); scores (NEM and the average between Language and Math); and the preference of assignment in the initial year. Odd columns include the entire sample, while even columns restrict the sample to students with a GPA greater than or equal to 4.0. Since 4.0 is the pass/fail threshold (the scale is from 1.0 to 7.0), the latter rules out forced switches.<sup>3</sup>

We observe that the magnitude of the signal is negatively correlated with students' likelihood of reapplying, modifying the composition of their rank-ordered list, and switching programs. Moreover, the signal is negatively correlated with the probability of switching

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<sup>3</sup>Our measure of GPA is aggregated, so it may not capture forced switches due to a single course (e.g., failing a mandatory class multiple times). However, students with high GPA who are forced to quit are uncommon.

TABLE II  
EFFECT OF SIGNAL ON REAPPLICATION AND SWITCHING

|                           | Reapplied            |                   | Changed comp.        |                  | Switch               |                     | Switch Down          |                      | Switch Up           |                     |
|---------------------------|----------------------|-------------------|----------------------|------------------|----------------------|---------------------|----------------------|----------------------|---------------------|---------------------|
|                           | (1)                  | (2)               | (3)                  | (4)              | (5)                  | (6)                 | (7)                  | (8)                  | (9)                 | (10)                |
| Signal                    | -0.980***<br>(0.033) | -0.092<br>(0.061) | -0.850***<br>(0.035) | 0.077<br>(0.073) | -0.790***<br>(0.031) | 0.226***<br>(0.057) | -0.883***<br>(0.054) | -0.705***<br>(0.144) | 0.753***<br>(0.088) | 0.997***<br>(0.107) |
| Mean                      | 0.119                | 0.085             | 0.08                 | 0.057            | 0.131                | 0.099               | 0.022                | 0.015                | 0.024               | 0.026               |
| Test                      | 0.11***              | 0.086***          | 0.074***             | 0.074***         | 0.124***             | 0.093***            | 0.019***             | 0.017***             | 0.019***            | 0.018***            |
| Demographics              | Yes                  | Yes               | Yes                  | Yes              | Yes                  | Yes                 | Yes                  | Yes                  | Yes                 | Yes                 |
| Passed ( $GPA \geq 4.0$ ) | No                   | Yes               | No                   | Yes              | No                   | Yes                 | No                   | Yes                  | No                  | Yes                 |
| Observations              | 14,395               | 12,591            | 14,395               | 12,591           | 14,395               | 12,591              | 14,395               | 12,591               | 14,395              | 12,591              |

*Note:* Sample includes all students from the cohort that graduated high school in 2014, enrolled in 2014 and 2015 in a program offered by a university within the system, and for whom first-year grade data is available. GPA is measured on a 1–7 scale, with failing grades defined as below 4.0. Test reports the results of t-test assuming no signal (all p-values are below  $1 \times 10^{-4}$ ), i.e., testing whether the intercept is different than zero. Significance levels: \* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

to less preferred programs (“switching down”) and positively correlated with switching to more preferred programs (“switching up”). These patterns suggest that students who receive stronger signals—indicating they performed worst than expected and thus revise their beliefs about their unobserved ability downward—are less inclined to pursue more desirable programs and more inclined to downgrade their choices. These findings indicate that students respond to signals in ways consistent with learning about ability.

While signals about ability play a central role in shaping students’ decisions, our findings also suggest that students may learn about their preferences through their college experience. To illustrate this, Table II reports the results of t-tests on the predicted probabilities of reapplying, modifying the rank-ordered list (ROL), and switching behaviors, conditional on receiving a null signal—i.e., when actual grades are as expected. We find that even in the absence of a signal, students remain significantly likely to reapply, modify the composition of their ROL, and switch programs. This pattern is further supported by the summary statistics in Table III, which show that students who receive a neutral signal still engage in these behaviors at meaningful rates—generally falling between those observed for students receiving positive or negative signals. For instance, 8.6% of students with a neutral signal

reapply, compared to 19.9% of those with a positive signal and 7.6% of those with a negative signal. Similarly, 6.0% of students with a neutral signal change the composition of their ROL, versus 13.5% for positive and 5.3% for negative signals.

TABLE III  
SWITCHING STATISTICS BY SIGNAL TYPE

|          | N    | Reapply          |                  | Switch           |                  | Switch Down      |                  | Up               |
|----------|------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|          |      | All              | Changed comp.    | Program          | Major            | Program          | Major            | Program          |
| Negative | 5745 | 0.199<br>(0.005) | 0.135<br>(0.005) | 0.216<br>(0.005) | 0.098<br>(0.004) | 0.039<br>(0.039) | 0.014<br>(0.002) | 0.011<br>(0.001) |
| Neutral  | 1665 | 0.086<br>(0.007) | 0.060<br>(0.006) | 0.100<br>(0.007) | 0.043<br>(0.005) | 0.019<br>(0.019) | 0.007<br>(0.002) | 0.016<br>(0.003) |
| Positive | 8325 | 0.076<br>(0.003) | 0.053<br>(0.002) | 0.105<br>(0.003) | 0.041<br>(0.002) | 0.012<br>(0.012) | 0.003<br>(0.001) | 0.033<br>(0.002) |

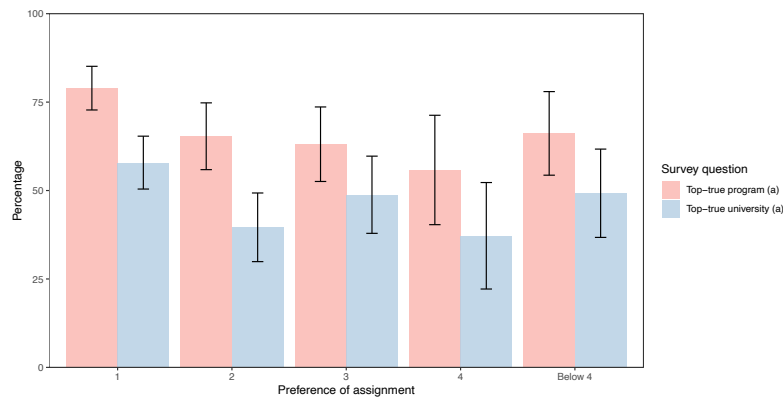
*Note:* Sample includes all students from the cohort that graduated high school in 2014, enrolled in 2014 in a program offered by a university within the system, and for whom first-year grade data is available. Positive / Negative signal considers students who received a sufficiently positive / negative signal, i.e., 0.1 points below or above expected, respectively. Neutral signal includes all students who did not receive a positive or negative signal.

Our previous results show that students' reported preferences may change during their first year in college. To rule out the possibility that these changes are driven by students' being more experienced in participating in the system (as opposed to them learning about their ability or preferences), we analyze changes in true preferences using the surveys conducted in 2019 and 2020. Specifically, we construct a panel that includes students who participated in both surveys and responded to the same questions (close to 1,300 students), and we compare the top true preference reported in each year. In Figure 3, we plot the fraction of students who changed their top true preference (for programs and also for universities) as a function of their initial preference of assignment (in 2019). First, we observe that on average, close to 65% of the students in the data changed their top true preference after their first year in college. Moreover, close to 50% of students even changed their most preferred university. Second, we observe that students initially assigned to lower preferences are less likely to change their top true preference for programs. This pattern is consistent with the existence of both the *mismatch* and *learning* channels: students assigned to lower-ranked



programs are likely affected by initial mismatches and, thus, a higher proportion keeps their initial top preference. In contrast, students initially assigned to their top-reported preference have a lower probability of being *mismatched*; thus, their reapplication suggests that they learned about their ability during their first year in college.

FIGURE 3.—Percentage of reapplicants that change their top-true preference, by preference of assignment in 2019



#### 4. MODEL

In this section, we present a dynamic model of students' application behavior and academic progression that incorporates how they learn about their abilities and preferences over time. The model adapts and extends the work from [Arcidiacono \(2004, 2005\)](#), [Arcidiacono et al. \(2025\)](#) to (i) capture the empirical patterns documented in the previous section; (ii) quantify the extent to which students' switching behavior is driven by initial mismatch versus subsequent learning; and (iii) assess how alternative learning environments and market design interventions affect students' outcomes in equilibrium.

*Overview.* We assume that students enter college with imperfect information about two key dimensions: (i) their match-specific abilities—as multidimensional skills relevant to labor market returns—and (ii) their preferences over majors of study. Abilities are partially revealed through college performance, while preferences evolve based on students' experience in their enrolled programs, independently of grades. This learning process shapes

students' beliefs about future returns upon graduation and informs their decisions to continue in their current program, reapply to the centralized system, or drop out. Finally, upon graduation, students enter the labor market and receive utility that depends both on a pecuniary component corresponding to their lifetime labor market returns and a non-pecuniary one that captures the alignment between their program of graduation and their preferences.

*Notation.* We use  $i \in \mathcal{I}$  to refer to students,  $j \in \mathcal{J} \cup \{\emptyset\}$  to refer to programs ( $\emptyset$  representing no enrollment), and  $t \in \mathcal{T} = \{1, \dots, T\}$  to refer to periods. In addition, let  $\mathcal{M}$  be the set of majors and  $m : \mathcal{J} \rightarrow \mathcal{M}$  be the function that maps programs to majors, i.e.,  $m(j)$  represents the major of program  $j$ . Finally, with a slight abuse of notation, we use functions to make explicit the underlying arguments and structure from which key elements of the model are derived (e.g.,  $G(\cdot)$  for grades); subscripts to denote their realized values (e.g.,  $G_{i,j,t}$ ); tilde notation to denote beliefs over these values (e.g.,  $\tilde{G}_{i,j,t}$ ); and hat notation to represent their estimated values (e.g.,  $\hat{G}_{i,j,t}$ ).

*Organization.* Section 4.1 describes the learning process during college. Section 4.2 specifies the flow utility from enrollment. Section 4.3 presents the workforce utility obtained upon graduation. Finally, Section 4.4 discusses implementation details, including timing, decisions, sources of heterogeneity, additional details considered in the estimation.

#### 4.1. Learning

As previously discussed, we assume that students learn about (i) their preferences through their enrollment decisions, and (ii) their unknown abilities through their grades.

*Learning about preferences.* We allow students to learn about their preferences for majors. In particular, we assume that student  $i$ 's idiosyncratic preferences for majors in period  $t$  are captured by a set of random coefficients  $\{\alpha_{i,m,t}\}_{m \in \mathcal{M}}$ , whose initial values are normally distributed with mean  $\mu_{i,m}^\alpha$  and standard deviation  $\sigma_{i,m}^\alpha$ . Furthermore, we assume that students only learn about their preferences for their major of enrollment. Thus, student  $i$ 's idiosyncratic preferences for major  $m$ , conditional on being enrolled in program  $j$ , evolves according to  $\alpha_{i,m,t+1} = \alpha_{i,m,t} + \mathbb{1}_{\{m=m(j)\}} \cdot \epsilon_{i,m,t}^\alpha$ , where  $\epsilon_{i,m,t}^\alpha \sim N\left(0, \left(\zeta_{i,m}^\alpha\right)^2\right)$ . Fi-

nally, as we later discuss in Section 4.2, students observe preference shocks (related to their flow utility) that are not persistent over time and i.i.d. across periods, so we do not consider these as part of the (systematic) learning process.

*Learning about abilities.* Beyond learning about their preferences, we allow students to learn about their abilities. Specifically, we assume that the ability  $A_{i,j}$  of student  $i$  in program  $j$  corresponds to the sum of three elements: (i)  $A_{i,j}^o$ , an observable component that is known by the student and the econometrician; (ii)  $A_{i,j}^p$ , a private component only known by the student; and (iii)  $A_{i,j}^u$ , unknown by both the student and the econometrician but learned from the grades obtained during college. More specifically, let  $\mathcal{S}$  be the set of subjects that are mandatory for all students in the standardized exam (i.e.,  $\mathcal{S} = \{s_m, s_v\}$ , with  $s_m$  and  $s_v$  representing the math and verbal parts of the standardized test, respectively),<sup>4</sup> and let  $\omega_{j,s}$  be the weight that program  $j$  places on subject  $s \in \mathcal{S}$ . Then, student  $i$ 's ability in program  $j$  is given by

$$A_{ij} = \underbrace{\sum_{s \in \mathcal{S}} \omega_{j,s} A_{i,s}^o}_{:= A_{i,j}^o} + \underbrace{A_{i,m(j)}^p}_{:= A_{i,j}^p} + \underbrace{\sum_{s \in \mathcal{S}} \omega_{j,s} A_{i,s}^u}_{:= A_{i,j}^u}. \quad (1)$$

Note that the private and unobserved components comprise two elements: (i)  $A_{i,m(j)}^l$ , which is major-specific, and (ii)  $A_{i,s}^l$ , which are subject-specific, with  $l \in \{p, u\}$ . For private abilities, we assume that these are randomly drawn at the beginning of the academic progression, with  $A_{i,k}^p \sim N(\mu_{i,k}^p, \varsigma_{i,k}^2)$  for  $k \in \mathcal{M} \cup \mathcal{S}$ , where  $\mu_{i,k}^p$  is the mean—normalized to zero for majors—and  $\varsigma_{i,k}$  is the standard deviation of student  $i$ 's private ability in program  $j$  and component  $k$ . For unobserved abilities, we assume that the true distribution of  $A_{i,k}^u$  is normal with mean zero and standard deviation  $\sigma_{i,k}^2$  for  $k \in \mathcal{M} \cup \mathcal{S}$ , but we allow students to have subjective, heterogeneous and potentially biased prior beliefs about these elements; namely, we assume that student  $i$ 's prior beliefs about  $A_{i,k}^u$  in period  $t$  is normally dis-

<sup>4</sup>Elective test scores (Science/History) and high school performance measures (NEM/Rank) are excluded from the ability term because they are not mandatory and consistently available for all students. However, elective scores are updated for exam retakers, and all five admission factors are used when estimating admission probabilities.

tributed with mean  $\tilde{\mu}_{i,k,t}$  and variance  $\tilde{\sigma}_{i,k,t}^2$ , with initial values  $\tilde{\mu}_{i,k}$  and  $\tilde{\sigma}_{i,k}^2$ . Consequently, student  $i$ 's prior mean and variance about their unobserved ability in program  $j$  in period  $t$  are given by  $\tilde{\mu}_{i,j,t} = \tilde{\mu}_{i,m(j),t} + \sum_{s \in \mathcal{S}} \omega_{j,s} \cdot \tilde{\mu}_{i,s,t}$  and  $\tilde{\sigma}_{i,j,t}^2 = \tilde{\sigma}_{i,m(j),t}^2 + \sum_{s \in \mathcal{S}} \omega_{j,s} \cdot \tilde{\sigma}_{i,s,t}^2$  respectively. Furthermore, we assume that students' grades are a function of a set of time invariant observable characteristics  $X_{i,j}^G$ — which include characteristics of student  $i$  (e.g., their observed ability) and program  $j$  (e.g., major)—, their private and unknown abilities  $A_{i,j}^p, A_{i,j}^u$ , and a noise term  $\epsilon_{i,j,t}^G \sim N(0, \sigma_G^2)$ , i.e.,

$$G_{i,j,t} = G(X_{i,j}^G, A_{i,j}^p, A_{i,j}^u, \epsilon_{i,j,t}^G). \quad (2)$$

Consequently, based on their prior beliefs about their unobserved ability, students construct beliefs about their grades, which we denote by  $\tilde{G}_{i,j,t} := \mathbb{E}[G_{i,j,t}]$ —the expectation is computed over  $\tilde{A}_{i,j}^u \sim N(\tilde{\mu}_{i,j,t}, \tilde{\sigma}_{i,j,t}^2)$  and  $\epsilon_{i,j,t}^G \sim N(0, \sigma_G^2)$ —and, upon observing their actual grades, they receive a noisy signal that they use to update their prior beliefs about their unknown abilities according to Bayes rule, as formalized in Proposition 1. We defer the proof to Appendix C.1.

**PROPOSITION 1:** *Let  $\{\tilde{\mu}_{i,j',t}, \tilde{\sigma}_{i,j',t}\}_{j' \in \mathcal{J}}$  be the sets of prior means and variances of student  $i$ 's unobserved ability in programs  $j' \in \mathcal{J}$  at time  $t$ , respectively. In addition, let  $j \in \mathcal{J} \cup \{\emptyset\}$  be the program of enrollment of student  $i$  in period  $t$ , and  $a_{i,j,t}$  be the observed signal (if  $j \in \mathcal{J}$ ). Then, the posterior mean of student  $i$  in program  $j' \in \mathcal{J}$  is given by*

$$\tilde{\mu}_{i,j',t+1} = \begin{cases} \tilde{\mu}_{i,j',t} + \frac{(a_{i,j,t} - \tilde{\mu}_{i,j,t})}{\tilde{\sigma}_{i,j,t}^2 + \sigma_G^2} \cdot \left[ \tilde{\sigma}_{i,m(j),t}^2 + \sum_{s \in \mathcal{S}} \omega_{j's} \omega_{js} \tilde{\sigma}_{i,s,t}^2 \right] & \text{if } m(j') = m(j), j \in \mathcal{J} \\ \tilde{\mu}_{i,j',t} + \frac{(a_{i,j,t} - \tilde{\mu}_{i,j,t})}{\tilde{\sigma}_{i,j,t}^2 + \sigma_G^2} \cdot \left[ \sum_{s \in \mathcal{S}} \omega_{j's} \omega_{js} \tilde{\sigma}_{i,s,t}^2 \right] & \text{if } m(j') \neq m(j), j \in \mathcal{J} \\ \tilde{\mu}_{i,j',t} & \text{if } j \notin \mathcal{J} \end{cases}$$

Intuitively, students learn more about programs that resemble the one they are currently enrolled in—particularly those within the same major or with comparable weighting of admissions scores. Finally, we assume that students' beliefs are heterogeneous across students; in Table C.II in Appendix C.6, we show how these vary among several observable characteristics, including students' gender, income level, and observed ability.

## 4.2. Flow Utility

Let  $U_{i,j,t}$  be the flow utility that student  $i$  receives for attending program  $j$  in period  $t$ . We assume that this utility can be modeled as:

$$U_{i,j,t} = U \left( X_{i,j}^U, \alpha_{i,m(j),t}, \epsilon_{i,j,t}^U \right), \quad (3)$$

where  $X_{i,j}^U$  is a set of time-invariant observable characteristics that include characteristics of student  $i$  (e.g., gender, income, or observed ability), program  $j$  (e.g., fixed effects, quality, or selectivity), and interaction terms (e.g., the distance between student  $i$ 's and program  $j$ 's municipalities or the net cost that student  $i$  pays for program  $j$ );  $\alpha_{i,m(j),t}$  is student  $i$ 's idiosyncratic preference for major  $m(j)$  in period  $t$ ; and  $\epsilon_{i,j,t}^U$  is an idiosyncratic preference shock that is distributed i.i.d across programs and periods and follows a type I extreme value distribution with a scale parameter  $\kappa$ . We specify a location normalization and set the systematic value of the outside option to  $U_{i,\emptyset,t} = 0$  for all  $t \in \mathcal{T}$  and set the distance coefficient to minus one, allowing the scale of the shocks to vary.

## 4.3. Labor Market

Upon graduating from program  $j \in \mathcal{J} \cup \{\emptyset\}$  in period  $t' \in \mathcal{T}$ , we assume that student  $i$  enters the work force and obtains a utility  $V_{i,j,t'}$  consisting of two elements: (i) a non-pecuniary utility  $V_{i,j,t'}^{NP}$ , capturing the utility derived from their preferences during their time in college; and (ii) a pecuniary utility  $V_{i,j,t'}^P$ , capturing the present value of their lifetime wages. Given a retirement period  $T$  and a common discount factor  $\delta$ , we model this as:

$$V_{i,j,t'} = \underbrace{\beta_{m(j)}^V + \beta_{\alpha}^V \alpha_{i,m(j),t'} + \beta_A^V A_{i,j}^o + \beta_{PU}^V \cdot (A_{i,j}^p + A_{i,j}^u)}_{:=V_{i,j,t'}^{NP}} + \underbrace{\beta_W^V \log \left( \mathbb{E} \left[ \sum_{t=t'}^T \delta^{t-t'} \cdot P_{m(j),t}^W \cdot W_{i,j,t} \right] \right)}_{:=V_{i,j,t'}^P}. \quad (4)$$

The first term captures the non-pecuniary utility, which depends on student  $i$ 's preferences for major  $m(j)$  encoded in the random coefficient  $\alpha_{i,m(j),t'}$  and on their abilities. The second term captures the pecuniary utility, which is separable over time and depends on the

(major-specific) employment probability  $P_{m(j),t}^W$  and the wage  $W_{i,j,t}$ , which we model as

$$W_{i,j,t} = W \left( X_{i,j}^W, G_{i,j,t'}, \Lambda_{j,t-t'} \right), \quad (5)$$

where  $X_{i,j}^W$  is a set of time-invariant observable characteristics of student  $i$ ; program  $j$  and their interaction;  $G_{i,j,t'}$  is student  $i$ 's GPA at the time of graduation; and  $\Lambda_{j,t-t'} = \Lambda(X_{i,j}^\Lambda, t - t')$  captures the (potentially heterogeneous across observable characteristics in  $X_{i,j}^\Lambda$ ) effect of tenure on program  $j$ 's wage. As we discuss later, we assume that the effect of grades on wages is heterogeneous across students, and we model it as a random coefficient that is normally distributed with mean  $\mu_{W \sim G}$  and variance  $\sigma_{W \sim G}^2$ . Finally, note that student  $i$  receives this continuation value only if they graduate from their program. If, instead, student  $i$  drops out, we assume that their continuation value is equal to  $V_{i,\emptyset,t'}$ .

#### 4.4. Implementation

*Timing.* For estimation, we consider a three-period model. Periods 1 and 2 correspond to the first and second years of college, while period 3 begins in the third year and aggregates all subsequent years until graduation, including discounted labor market payoffs. In period 1, students apply to colleges, receive their assignment, decide whether to retake the PSU, obtain first-year grades, and update their beliefs. In period 2, they may reapply and, given their current enrollment, new assignment, and what they learned earlier, decide whether to remain, switch programs, or drop out. In period 3, students face dropout and graduation probabilities and enter the labor market.

*Decisions.* Throughout their first two years in college, the key decisions we model are (i) enrollment (in period  $t = 2$ ), i.e., conditional on the assignment, whether to enroll in the newly assigned program, stay in the current one, or dropout; (ii) exam re-taking (in period  $t = 1$ ), i.e., whether or not to re-take the standardized test; and (iii) application (in periods  $t \in \{1, 2\}$ ), i.e., conditional on the scores received after (re)taking the test, the current beliefs about abilities and the updated preferences, which programs to include in the ROL (if any). For the latter, we assume that students differ in their level of sophistication ([Pathak and Sönmez, 2008](#)), with each student's application strategy being exogenously given and

following one of two types: (i) weak truth-telling (w.p.  $\rho$ ), i.e., students report their true preferences; and (ii) strategic (w.p.  $1 - \rho$ ), i.e., students submit a ROL that maximizes their expected value (as in [Chade and Smith \(2006\)](#)) assuming correct beliefs about their marginal admission probabilities. Specifically, a strategic student  $i$  solves:

$$R_i^* \in \operatorname{argmax}_{R \subseteq \mathcal{R}_i} \left\{ \sum_{l=1}^{|R|} p_{i,R(l),t} \cdot \bar{p}_{i,R(l),t} \cdot \Psi_{i,R(l)}^* - C_{i,t}^A(R) \right\}, \quad (6)$$

where  $p_{i,j,t}$  is the probability of getting assigned to program  $j$  in period  $t$ ;  $\bar{p}_{i,R(l),t} = \prod_{l'=1}^{l-1} (1 - p_{i,R(l'),t})$  is the probability of not being assigned to any of the top  $l - 1$  programs listed in  $R$ , with  $\bar{p}_{i,R(1),t} = 1$ ;  $C_{i,t}^A(R) = c|R|$  is the cost of applying to ROL  $R$ ; and  $\mathcal{R}_i := \{R \in \mathcal{R}(\mathcal{J}) : |R| \leq K, p_{i,j',1} > 0, \forall j' \in R\}$  is the set of ROLs involving at most  $K$  programs with positive admission probability. Note that students include programs in their ROL only if it is strictly profitable, so they will skip programs for which their admission chances are zero. This assumption is in line with the findings in ([Larroucau et al., 2025](#)), and rules out multiplicity of optimal ROLs involving zero-chance programs ([He, 2012](#)).

The key element in (6) is the term  $\Psi_{i,R(l)}^*$ , which represents the expected continuation value associated with being assigned to program  $j = R(l)$  in period  $t$ . This value integrates all subsequent dynamic decisions that follow, including whether to enroll, retake the admissions exam, re-apply to new programs, switch programs or dropout and join the labor force (see Appendix C.2). It also embeds the evolution of beliefs about abilities and the learning processes that unfold conditional on enrollment—students update their expectations based on observed outcomes and experiences—and also capture the probabilities of graduation, dropout, which we model as functions of observable characteristics from period three (see Appendix C.3), and labor market outcomes upon graduation.

Computing these continuation values is particularly challenging, as it requires taking expectations over a high-dimensional state space that encompasses the multiple sources of uncertainty previously mentioned. To address this complexity, we assume a large matching market, so students take the distribution of cutoffs as given ([Azevedo and Leshno, 2016](#)), and we adapt the results of [Fack et al. \(2019\)](#) to our dynamic setting, showing that students' continuation values over their optimal portfolios can be represented by their most

preferred ex-post feasible programs, integrated over the empirical distribution of future cutoffs. Exploiting the assumption that preference shocks are i.i.d. Type-I extreme value, we then approximate the corresponding expectations using a log-sum expression (see Appendix C.4). This approach allows us to evaluate continuation values without integrating over all preference shocks, greatly reducing computational burden.

Finally, these continuation values can be directly influenced by policy interventions—for example, by altering the information environment, which shapes the learning process, or through market design changes that modify the incentives and feasibility of re-application or switching. We discuss such interventions in greater detail in Section 7.

*Heterogeneity.* The learning process described above incorporates heterogeneity at the student level, allowing the learning parameters—related to private signals ( $\varsigma_{i,k}$  for  $k \in \mathcal{M} \cup \mathcal{S}$ ), unknown abilities ( $\sigma_{i,k}$ ,  $\tilde{\mu}_{i,k}$ ,  $\tilde{\sigma}_{i,k}$  for  $k \in \mathcal{M} \cup \mathcal{S}$ ), and preferences ( $\mu_{i,m}^\alpha$ ,  $\sigma_{i,m}^\alpha$ , and  $\zeta_{i,m}^\alpha$  for  $m \in \mathcal{M}$ )—to vary across students and majors/subjects. To ease estimation, we group majors into four *broad major*—Science (Science, Farming, and Technology), Social Sciences (Social Sciences, Business and Management, Art and Architecture, and Law), Education and Humanities, and Health. Major-specific structural parameters vary at the broad major level, while random variables (e.g., major-specific abilities) vary at the *major* level. We also classify programs as *math intensive* (*verbal intensive*) if their math weight exceeds the verbal one (otherwise) and use the average math weight within each type, i.e.,  $\frac{\sum_{j \in \mathcal{J}} \omega_{j,sm} \cdot \mathbb{1}_{\{s(j)=sm\}}}{\sum_{j \in \mathcal{J}} \mathbb{1}_{\{s(j)=sm\}}}$ , where  $s(j)$  indicates the type of program  $j$ . Then, the unobserved ability of student  $i$  in program  $j$  is  $A_{ij}^u = A_{i,m(j)}^u + \bar{\omega}_{s(j)} A_{i,sm}^u + (1 - \bar{\omega}_{s(j)}) A_{i,sv}^u$ .

Furthermore, we capture heterogeneity at the student-level using a small set of observable characteristics that are likely to influence the learning process and biases in students' beliefs. Specifically, we model each of the mean parameters mentioned above (i.e.,  $\tilde{\mu}_{i,k}$  for  $k \in \mathcal{M} \cup \mathcal{S}$  and  $\mu_{i,m}^\alpha$  for  $m \in \mathcal{M}$ ) as linear functions of gender, income level, and their average ability, measured as the normalized average between the Math and Verbal entrance exams. We apply a similar approach to the logarithm of the variance parameters ( $\varsigma_{i,k}$ ,  $\sigma_{i,k}$  for  $k \in \mathcal{M} \cup \mathcal{S}$  and  $\sigma_{i,m}^\alpha$ ,  $\zeta_{i,m}^\alpha$  for  $m \in \mathcal{M}$ ). For instance, for the true variance of unknown abilities, we assume  $\log(\sigma_{i,k}^2) = \gamma_k^\sigma + \gamma_F^\sigma \cdot \text{female}_i + \gamma_{LI}^\sigma \cdot \text{low-income}_i + \gamma_A^\sigma \cdot \bar{A}_i$ , where



female<sub>*i*</sub>, low-income<sub>*i*</sub> are dummy variables for whether student *i* is female or their income below the median, respectively, and  $\bar{A}_i = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} A_{i,s}^o$  is *i*'s average observable ability.<sup>5</sup>

*Functional forms.* We assume that the grade, flow utility, and log-wage functions (in (2), (3) and (5), respectively) are linear in their arguments, i.e.,

$$G\left(X_{i,j}^G, A_{i,j}^p, A_{i,j}^u, \epsilon_{i,j,t}^G\right) = \beta_X^G X_{i,j}^G + A_{i,j}^p + A_{i,j}^u + \epsilon_{i,j,t}^G, \quad (7a)$$

$$U\left(X_{i,j}^U, \alpha_{i,m(j),t}, \epsilon_{i,j,t}^U\right) = \beta_X^U X_{i,j}^U + \alpha_{i,m(j),t} + \epsilon_{i,j,t}^U, \quad (7b)$$

$$\log W\left(X_{i,j}^W, G_{i,j,t'}, \Lambda_{j,t-t'}\right) = \beta_X^W X_{i,j}^W + \beta_{G,i}^W G_{i,j,t'} + \Lambda_{m(j),t-t'} + \epsilon_{i,j,t}^W, \quad (7c)$$

where  $\Lambda_{m(j),t-t'} = \beta_{m(j),1}^\Lambda (t - t') + \beta_{m(j),2}^\Lambda (t - t')^2$ . Furthermore, following our survey evidence, we assume that the effect of grades on wages,  $\beta_{G,i}^W$ , are heterogeneous across students and can be modeled as a random coefficient that is log-normally distributed with mean  $\log(\mu_{W \sim G})$  and variance  $\sigma_{W \sim G}^2$ . Finally, note that for estimation, we use beliefs on expected wages and grades as opposed to their actual values, so we estimate (7c) considering  $\tilde{W}_{i,j,t}$ ,  $\tilde{G}_{i,j,t}$  and adding a measurement error term. This approach, made possible by our rich survey data, allows us to relax the rational expectations assumption and enable us to more accurately capture the (potentially biased) information that students actually used when making their decisions (Wiswall and Zafar, 2015a,b). Furthermore, it allows us to understand the role of biases (in both mean and variances) in affecting the dynamics of college admissions and how different interventions can mitigate these.

*Controls.* As shown in (7), each function controls for a set of time-invariant observable characteristics—encoded in  $X_{i,j}^G$ ,  $X_{i,j}^U$  and  $X_{i,j}^W$ —that include characteristics of the student, the program, and their interaction. In the grade equation, we control for broad major fixed effects and students' observed ability. The flow-utility equation includes program fixed effects—parametrized as a combination of major fixed effects, college fixed effects, a quality index and dummies for the top 20 programs in terms of market shares—, observed

<sup>5</sup>We use the average instead of the program-specific observed ability (multiplying each  $A_{i,s}^o$  by the corresponding weight  $\omega_{j,s}$ ) to avoid program-specific heterogeneity in defining students' types and simplify the estimation.

ability, distance to the program's municipality, distance to the program's mean quality, and net costs (tuition minus scholarships). Finally, the wage equation includes broad major fixed effects, expected grades at graduation, average posted wages at enrollment, and fixed effects for gender and low-income status. See Appendix C.5 for additional details.

## 5. IDENTIFICATION

We outline our identification strategy by focusing on the sources of variation that identify the main parameters. While full identification relies on all moment conditions, several parameters can be isolated using targeted moments. For clarity, we present the arguments constructively and conditional on observable characteristics.

*Preferences.* We follow Agarwal and Somaini (2018), who show that students' preferences can be non-parametrically identified from their reported ROLs assuming correct beliefs about their (marginal) admission probabilities and exploiting distance as a special regressor.

*Application.* To estimate the probability of being weak truth-teller in the baseline ( $\rho$ ), we use (i) the share of applications for which the top-reported choice has zero admission probability, and (ii) the share of applicants who include their top-true preference (elicited in the survey) as their top-reported program. The first moment helps to identify strategic behavior by capturing applications to programs with no admission chance. The second moment directly measures truth-telling alignment.

*Tuition and welfare effects.* We follow Kapor et al. (2024) and exploit the discontinuous change in tuition generated by the scholarship *Beca Vocación de Profesor* and its heterogeneous effect on enrollment to identify the effect of the net cost that students pay.<sup>6</sup>

*Private abilities and preferences.* Students know their private abilities and the draw of their initial preferences when making their initial choices (Arcidiacono, 2005). Thus, we

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<sup>6</sup>Students with average scores above 600 can enroll in Education programs tuition-free, creating a discontinuity in enrollment around this cutoff (see Figure D.1 in Appendix D.2) that we exploit for identification.

exploit the correlation in students' initial reported preferences—i.e., before learning about their unknown abilities—across majors and subjects to identify the role of private information on students' choices, and rely on the correlation between students' ROL composition and first-year grades to identify the effect of private abilities on grades using a control function approach (Heckman, 1979, Kline and Walters, 2016). If private abilities largely shape preferences through grades, we should observe a strong correlation between the two.

*Variance of abilities and grades.* Since variation in students' grades, can only be explained by their private abilities, their unknown abilities, and the grade shock—the former already identified—, we separately identify the variance of unknown abilities from the variance of the grade noise by constructing a system of equations using the variance of the first-year and second-year signals among non-switchers (see Appendix D.1).

*Biased unknown abilities.* Assuming that beliefs about grades follow a similar structure to that in Equation (7a), considering students' (potentially biased) prior about their unknown ability and an additional measurement error term  $\epsilon_{i,j,t}^{\tilde{G}} \sim N(0, \sigma_{\tilde{G}}^2)$ , we identify the initial mean bias in students' beliefs about their first-year grade,  $\tilde{\mu}_{i,j}$ , by averaging the difference between students' beliefs about their first-year grades and their actual grades.

*Learning about preferences and abilities.* We directly identify subjective prior variances exploiting the probability that students assign to their grades falling between their expected grades plus or minus 10 percentiles conditional on attending their *top-reported program*, which reflect students' uncertainty about their GPA given their subjective beliefs. We also use the correlation between the signal and the choices that students make after the first period (including their decisions to switch and dropout) to further separate the effect of subjective beliefs from grade noise. Intuitively, a high signal-to-noise ratio implies a low variance of the grade shock relative to the subjective variance of unknown abilities, which means students' choices respond more strongly to their observed grades. This translates into stronger correlations between first-year grades and switching decisions. Moreover, if students are more likely to switch majors than subjects when they receive negative signals (lower grades than expected), then the variances related to majors outweigh those for

subjects. Finally, we identify the standard deviation of the evolution of random coefficients,  $\zeta_{i,m}^\alpha$ , by exploiting variation in preference changes across majors among students who reapply after receiving a non-informative signal, allowing us to isolate the effect of changing preferences from learning about unknown abilities.

*Labor Market.* To identify the (subjective) wage equation, we rely on the correlation between students beliefs about their wages and grades at the individual level and across programs, including students' reported preferences, their top-true program, and also a random program outside their reported ROL—ruling out potential selection concerns. This type of variation allows us to estimate the marginal effect of expected grades on wages. To separately identify the wage equation parameters from measurement error, we exploit the fact that we elicited students' beliefs about their admission chances to their top-reported program in two ways: (i) directly, and (ii) indirectly, by asking them to report the probability that their application score will be above the cutoff. The difference between these two responses should be independent of students unknown ability, providing a proxy of the measurement error in the survey. Then, to separate the variance of the effect of grades on wages from the measurement error—i.e.,  $\sigma_{W \sim G}^2$  from  $\sigma_{W \sim M}^2$ —we leverage the elicited beliefs on wages for different programs (both included and excluded from the reported preferences)—same as we do for beliefs in grades—to construct a system of equations involving the residuals of the subjective wage regression, as discussed in Appendix D. Finally, we identify the parameter scaling the wage component by leveraging the correlation between students' subjective wage beliefs and their preferences—specifically, by comparing top true and random programs.

*Initial mismatch and Counterfactual outcomes.* To identify the strength of the initial mismatch channel and its relationship to parameters governing unobserved heterogeneity and first-time enrollment costs, we exploit variation in initial assignments generated by the RDDs described in Section 3.2. In addition, these variations are key to identify the distribution of outcomes in the counterfactual analysis.<sup>7</sup> Indeed, because the primary effect of our

<sup>7</sup>See Agarwal et al. (2024a) for a thorough discussion of this identification challenge.

counterfactuals is to reassign students around admission cutoffs, these local variations provide a credible basis for predicting counterfactual outcomes. We then use the full model's structure to extrapolate beyond the cutoffs and account for equilibrium effects that may alter application behavior and initial assignments.

## 6. ESTIMATION

We estimate the model using a random sample of 4,000 students drawn from the population described in Section 3.1. We next describe the estimation procedure (Section 6.1), present the estimated parameters (Section 6.2), and assess the model fit (Section 6.3).

### 6.1. Estimation Procedure

*Estimation of Exogenous Components.* As discussed in Section 4.4, we estimate the distributions of admission probabilities  $\hat{p}_{j,t}$  using the bootstrap procedure detailed in Appendix C.3.1. Similarly, we estimate the probabilities of enrollment  $\hat{P}_{i,t}^e$ , graduation  $\hat{P}_{i,j,t}^g$  and dropout  $\hat{P}_{i,j,t}^d$  as discussed in Appendices C.3.2 and C.3.3, respectively. All these probabilities are estimated outside the main estimation procedure, serving as inputs.

*Indirect Inference.* We estimate the model parameters via Indirect Inference (II), which relies on building a statistical model that yields a rich description of the data patterns to identify the model parameters (Bruins et al., 2018). This statistical model—also known as the *auxiliary* model—is estimated on both the data and on simulated data from the structural model. The II estimator—also known as Wald approach to II—minimizes a function of the distance between the estimated data parameters and those estimated from the simulated data. In this sense, the Simulated Method of Moments is a particular case of II, in which the *auxiliary* model is a vector of moments. In Appendices E.1, E.2 and E.3, we formalize the estimator, describe the estimation algorithm, and discuss the moment conditions, auxiliary models and target parameters, respectively.

*Marginal Improvement Algorithm.* An important part of the estimation procedure is to solve for the optimal portfolio that a strategic student would choose, given the estimated parameters. To do this, we use the Marginal Improvement Algorithm (MIA) introduced

TABLE IV  
ESTIMATION RESULTS: SUMMARY OF KEY PARAMETERS

| Parameter                | Unknown Abilities      |      |                                |      |                        |      |                         |        |
|--------------------------|------------------------|------|--------------------------------|------|------------------------|------|-------------------------|--------|
|                          | Beliefs biases         |      |                                |      | True                   |      | Preferences             |        |
|                          | Mean ( $\tilde{\mu}$ ) |      | Log-Var ( $\tilde{\sigma}^2$ ) |      | Log-Var ( $\sigma^2$ ) |      | Log-Var ( $\zeta_m^2$ ) |        |
|                          | Value                  | S.E. | Value                          | S.E. | Value                  | S.E. | Value                   | S.E.   |
| Science                  | 0.63                   | (-)  | -1.28                          | 0.21 | -0.27                  | 0.19 | 0.57                    | 0.09   |
| Social Science           | 0.66                   | (-)  | -2.03                          | 0.13 | -0.20                  | 0.22 | 1.14                    | 0.01   |
| Education and Humanities | 0.36                   | (-)  | -0.74                          | 0.16 | -0.58                  | 0.26 | 0.48                    | < 0.01 |
| Health                   | 0.84                   | (-)  | -1.99                          | 0.09 | -1.44                  | 0.18 | 2.69                    | 0.02   |
| Low income               | 0.08                   | (-)  | -0.19                          | 0.21 | 0.12                   | 0.09 | 2.47                    | 0.02   |
| Female                   | -0.19                  | (-)  | 0.20                           | 0.25 | -0.37                  | 0.10 | -0.04                   | 0.02   |
| Avg. obs. ability        | -0.21                  | (-)  | -1.90                          | 0.08 | -0.80                  | 0.16 | 1.20                    | 0.03   |

Note: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator.

Parameters from external regressions are indicated by (-).

by Chade and Smith (2006), which sequentially adds the program that yields the highest marginal utility gain, stopping when the list-length constraint binds or no further improvement is possible. We then assume that a student's reported ROL corresponds to the MIA-selected programs, ordered by decreasing utility. See Appendix E.4 for details.

## 6.2. Estimated parameters

Table IV presents a summary of the key estimated parameters. Full estimation results are presented in Tables E.III, E.IV, E.V, E.VI, and E.VII in Appendix E.6.

Students' beliefs about their academic abilities display systematic biases. Prior means are upward-biased across all major categories, with subjective mean biases ( $\tilde{\mu}_m$ ) ranging from 0.36 to 0.84, consistent with pervasive overconfidence. Students also underestimate the uncertainty in their abilities, as indicated by negative biases in log-variances reflecting overprecision in their beliefs, i.e.,  $\log(\tilde{\sigma}_m^2) < \log(\sigma_m^2)$ . These distortions have important behavioral implications: overconfidence leads students to overestimate their chances of success in competitive programs, generating ex post mismatches once grade signals reveal true abilities; overprecision dampens belief updating in response to new information, slow-

ing learning and discouraging optimal switching. Biases also vary systematically across demographics: low-income students tend to be more overconfident and overprecise than high-income students; female students are generally less overconfident and overprecise than males; and higher-ability students are less overconfident but more overprecise.

Beyond learning about their abilities, students also learn about their preferences for majors through direct experience. The estimated log-variance parameters ( $\zeta^2$ ) reveal substantial heterogeneity across majors and student characteristics. Low-income students, for example, display higher variances and thus greater initial uncertainty about their preferences.

A few additional parameters are worth highlighting (see Appendix E.6 for details). The estimated share of strategic students is 0.88, suggesting that most applicants account for admission probabilities when choosing their rank-ordered lists, underscoring the importance of explicitly modeling strategic behavior. Regarding preference heterogeneity, the estimated scale of idiosyncratic shocks ( $\kappa = 17.78$ ) and the standard deviation of major-specific random coefficients ( $\sigma_\alpha = 33.7$ ) indicate substantial variation across students, with systematic differences in major preferences dominating idiosyncratic tastes. Finally, the variance of the grades shock ( $\sigma_g^2 = 0.11$ ) and the cross-sectional variance of the realized grade signal ( $\mathbb{V}(a_{i,j,1}) \approx 0.26$ ) imply that the grade signals are highly informative and, thus, students can meaningfully learn about their abilities from their performance in college.

Overall, these findings reveal that students exhibit substantial and heterogeneous belief biases—typically overestimating their abilities and underestimating uncertainty—yet the grade signal remains highly informative about their unknown abilities. These patterns can have far-reaching implications for application strategies, learning dynamics, and the welfare effects of policy interventions, highlighting the importance of accounting for both bias and heterogeneity in students' beliefs and learning parameters.

### 6.3. Model fit

To illustrate the key identifying variations discussed in Section 5, Table V summarizes the moments that help identify the two main channels: (i) initial mismatches and (ii) learning about abilities and preferences. Panel Va reports the estimated causal effects from the

TABLE V  
GOODNESS OF FIT: ADDITIONAL MOMENTS

| (a) RDD Moments           |        |        | (b) Beliefs on Prior Log-Variance |        |        |        |        |        |                |                                          | (c) Signal Correlations |        |      |
|---------------------------|--------|--------|-----------------------------------|--------|--------|--------|--------|--------|----------------|------------------------------------------|-------------------------|--------|------|
| Targets                   | Model  | Data   | Belief Variance                   |        |        |        |        |        | Learning Pref. |                                          | Targets                 | Model  | Data |
|                           |        |        | Subjective                        |        |        |        | Real   |        |                |                                          |                         |        |      |
|                           |        |        | Coefficient                       | Model  | Data   | Model  | Data   | Model  | Data           |                                          |                         |        |      |
| <i>RDD Switching</i>      |        |        | Science                           | -1.602 | -1.768 | -0.252 | -0.191 | 0.117  | 0.089          | <i>Grades <math>\times</math> Switch</i> |                         |        |      |
| Constant                  | 0.120  | 0.167  | Social Science                    | -1.709 | -1.746 | -0.227 | -0.139 | 0.053  | 0.029          | Dropout                                  | -0.220                  | -0.281 |      |
| Discontinuity             | -0.088 | -0.051 | Educ. and Human.                  | -1.522 | -1.770 | -0.480 | -0.548 | 0.080  | 0.048          | Program                                  | -0.158                  | -0.218 |      |
| <i>RDD Re-Application</i> |        |        | Health                            | -1.836 | -1.749 | -0.957 | -0.825 | 0.058  | 0.070          | Broad Major                              | -0.134                  | -0.158 |      |
| Constant                  | 0.276  | 0.234  | Female                            | -0.031 | -0.070 | -0.249 | -0.245 |        |                | Major                                    | -0.166                  | -0.213 |      |
| Discontinuity             | -0.144 | -0.087 | Low-income                        | 0.001  | 0.003  | 0.081  | 0.068  |        |                | Math Type                                | -0.117                  | -0.139 |      |
| <i>RDD Switch Up/Out</i>  |        |        | Observed ability                  | -0.307 | -0.302 | -0.524 | -0.503 |        |                | Switch Up                                | -0.027                  | 0.065  |      |
| Constant                  | 0.067  | 0.048  | Verbal weight <sup>2</sup>        | -0.069 | -0.175 | 0.092  | 0.351  |        |                | Switch Down                              | -0.048                  | -0.133 |      |
| Discontinuity             | -0.063 | -0.034 | Signal                            |        |        |        |        | -0.082 | -0.120         | Out (feas.)                              | -0.153                  | -0.225 |      |
|                           |        |        |                                   |        |        |        |        |        |                | Out (unfeas.)                            | -0.032                  | -0.011 |      |

Notes: Goodness of fit for key moments used in estimation. Panel (a) shows RDD discontinuity coefficients. Panel (b) shows regression coefficients of log total variance of prior beliefs (Belief Variance columns show both Model and Data for Subjective and Real beliefs) and learning preference coefficients (showing both Model and Data). Panel (c) shows correlations between grades and switching behaviors.

RDD models around admission cutoffs for students' top-reported preferences. These variations are crucial for identifying initial mismatches and accurately predicting switching rates in counterfactuals, since students may shift around the cutoffs in equilibrium. The model matches these moments reasonably well, though it slightly overpredicts the magnitudes of the discontinuities. Panel Vb shows the fit of regressions of the log total variance of students' prior beliefs on demographics and ability measures. Along with the correlation structure of grade signals, these moments identify the degree of learning about unknown abilities and their effects on outcomes. The model replicates well the observed heterogeneity across broad majors and demographic groups (see Table C.II in Appendix E.6). The same panel also reports the fit for regressions of changes in the norm of reported preferences on students' grade signals and major. While preference changes and switching are negatively correlated with grade signals (slope), there remains residual switching among students whose grades convey no information—evidence that students also learn about their match-quality (preferences) outside the grade production function. See Appendix E.3 for detailed specifications of these auxiliary models. Finally, Panel Vc presents the correlations between grade signals and switching outcomes. The model reproduces these patterns



1 closely: switching *Up* is nearly uncorrelated with grades, while switching *Out* to ex ante  
 2 feasible programs is strongly negatively correlated, consistent with both channels.

## 3 7. COUNTERFACTUALS 4

5 We now turn to our counterfactual analysis. As outlined in Section 1, we consider two  
 6 families of counterfactual policies: (i) changes to the information and learning environment,  
 7 aimed at reducing belief biases and front-loading learning about abilities and preferences;  
 8 (ii) market design interventions aimed to reduce initial mismatches and capture students'  
 9 dynamic incentives. We consider two types of market design interventions: (ii-a) modifica-  
 10 tions to the assignment mechanism designed to elicit cardinal preferences and mitigate ini-  
 11 tial mismatches; and (ii-b) adjustments to the reapplication rules that encourage students to  
 12 internalize the systemic effects of their dynamic decisions. We evaluate each of these poli-  
 13 cies in equilibrium, accounting for the strategic incentives they may generate—that is, we  
 14 assume that strategic students respond to changes in the environment by updating their be-  
 15 liefs about the distribution of cutoffs and adjusting their behavior accordingly. Algorithm 2  
 16 in Appendix F.3—which extends Kapor et al. (2020) to account for the mixture between  
 17 first-time and re-applicants—describes how we estimate students' equilibrium beliefs over  
 18 the stationary cutoff distributions.

19 Before proceeding to our primary counterfactual analysis, we assess how switching  
 20 and dropout decisions are shaped by the two behavioral channels (see Table F.I in Ap-  
 21 pendix F.5). First, eliminating biases about mean unknown abilities—by setting  $\tilde{\mu} = 0$ —  
 22 decreases switching by roughly 15%, while increasing unassignment by close to 19%. Sec-  
 23 ond, eliminating biases in the mean and variance of unknown abilities—by additionally  
 24 setting  $\tilde{\sigma}_k^2 = \sigma_k^2$  for  $k \in \mathcal{M} \cup \mathcal{S}$ —increases the number of switches and dropouts by close  
 25 to 81% and 23%, respectively. The intuition behind this result is that students are gener-  
 26 ally overoptimistic (i.e., positive mean bias) and underestimate variability (i.e., their prior  
 27 variance is lower than the true), so eliminating these biases increases the informativeness  
 28 of the signals received from grades, thereby increasing the value of choosing other options  
 29 (either switching or dropping out). Third, removing the learning about abilities channel—  
 30 by setting  $\sigma_k^2 = 0$  for  $k \in \mathcal{M} \cup \mathcal{S}$ —reduces program switching by roughly 23% relative

to the baseline. Beyond this, eliminating the learning about preferences channel—setting  $\zeta_{i,m}^\alpha = 0$  for all  $i \in \mathcal{I}$  and  $m \in \mathcal{M}$ —yields an additional 11% drop in switching and 12% in dropout. Finally, eliminating initial mismatches—by assigning every student to their top true choice—reduces switching by over 94%. Overall, the results indicate that initial mismatches and learning about abilities are the primary drivers of students’ dynamic decisions.

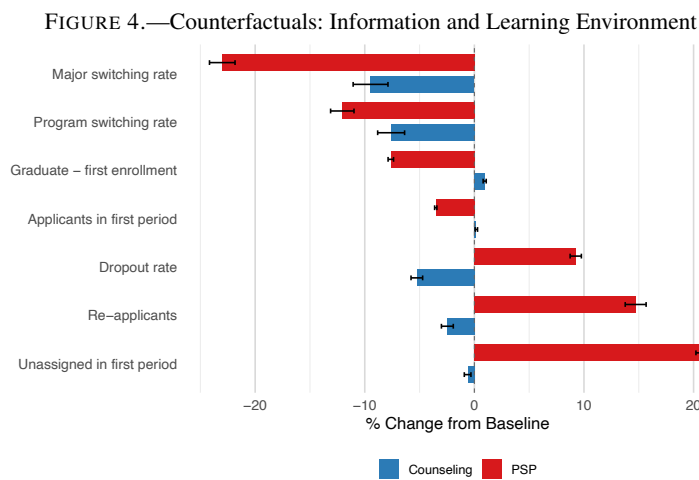
### 7.1. Information and Learning Environment

We first evaluate the effects of modifying the information environment to reduce belief biases and front-loading the learning process. The counterfactuals in this section are best interpreted as conceptual exercises designed—intentionally stylized to isolate the specific channels—to help us understand the mechanisms through which information and learning shape student outcomes in equilibrium, and to decompose the relative importance of learning about abilities versus learning about preferences. In particular, we consider two interventions:

1. Counseling: Students often receive (free) guidance and counseling during their last year of high school to help them make informed decisions about their future studies. To obtain an upper bound of their potential effect, we implement this counterfactual assuming that counseling eliminates the variance of the learning process about preferences, i.e.,  $\zeta_{i,m}^\alpha = 0$  for all  $i \in \mathcal{I}$  and  $m \in \mathcal{M}$ .
2. Mandatory year of Post Secondary Program (PSP): In some higher education systems (e.g., Canada and France), students are required to complete one- or multi-year preparatory programs before applying to a specific set of degree programs. This structure allows students to learn about their abilities prior to specialization without generating any crowd-out externality. We model this counterfactual as a one-year intervention implemented by: (i) debiasing the prior mean, i.e., setting  $\tilde{\mu}_{i,k} = 0$  for all  $i \in \mathcal{I}$  and  $k \in \mathcal{M} \cup \mathcal{S}$ ; (ii) transfer all the uncertainty from unknown abilities to private abilities, i.e., set  $\tilde{\sigma}_{i,k}^2 = 0$  and  $\hat{\zeta}_{i,k}^2 = \zeta_{i,k}^2 + \sigma_{i,k}^2$  for all  $i \in \mathcal{I}$  and  $k \in \mathcal{M} \cup \mathcal{S}$ ; and (iii) decreasing the terminal period by one year.

Both interventions seek to reduce inefficient switching by helping students make more informed application decisions, though they target different sources of learning. Counseling facilitates learning about students' preferences, allowing them to identify programs that align with their true interests before submitting their applications. In contrast, PSP enables students to learn about their abilities before applying to a specific program through the centralized system.<sup>8</sup>

*Results.* In Figure 4, we present the estimated effects—measured as percentage changes relative to the baseline—on several outcomes, including switching (program and major), dropout rates, unassigned and re-application rates, and graduation rates from the first program of enrollment. In Table F.II in Appendix F.5, we present the detailed results.



We observe that both interventions lead to a reduction in switching rates. However, the magnitude of some effects and the mechanism through which they operate differ significantly. First, we observe that the reduction in major switchings is significantly higher in the PSP intervention. The reason behind this result is that PSP allows students to fully learn

<sup>8</sup>Bordon and Fu (2015) analyze a shift from a joint system—where students apply directly to degree programs—to a sequential system, where students first choose a college and later declare a major. In contrast, our analysis modifies the learning environment leaving both the college structure and the assignment process unchanged.

about their major- and subject-specific unknown abilities, which play a more relevant role in driving students' dynamic decisions compared to learning about preferences. Second, we observe that PSP leads to a significantly higher number of unassigned students in the first period, which explains the higher incidence of re-applications compared to the counseling intervention. The reason behind this result is that debiasing prior means and eliminating the uncertainty about unknown abilities leads students to submit more "focused" rank-ordered lists, thereby increasing the risk of being unassigned. Finally, we observe that counseling leads to a reduction in first year and overall dropout rates, significantly improving the system's yield.

Overall, these results suggest that front-loading learning can help to reduce switching and dropout rates. However, the two interventions operate through different channels and have different implications. Counseling primarily helps students learn about their preferences, leading to a reduction in dropout rates, while a mandatory year of college allows students to learn about abilities—which have productive meaning—resulting in a more substantial decrease in switching rates, but also an increase in unassigned students.

## 7.2. Market Design Interventions

We now focus on market design interventions aimed at eliciting the intensity of students' preferences and accounting for their dynamic incentives. We consider two broad types of policy changes. The first targets the assignment mechanism, modifying how preferences are expressed and processed to better capture their intensity—either by restricting the length of applicants' rank-ordered lists (Constrained Deferred Acceptance, CDA) or enabling students to signal one of their preferences to receive a score bonus (Choice-Augmented Deferred Acceptance, CADA ([Abdulkadiroğlu et al., 2015](#))). The second set of interventions alters the reapplication rules, seeking to address dynamic incentives and improve the system's yield by adjusting how the mechanism treats first-time versus returning applicants, as in the Finnish and Turkish reapplication policies. In addition to the outcomes discussed in the previous section, we also evaluate the welfare implications of these policies. We add two measures of students' welfare: (i) interim and (ii) ex-post, which capture welfare be-

fore and after students learn about their abilities, respectively.<sup>9</sup> Both welfare measures are translated into millions of Chilean pesos as of 2014. Unlike before, we can now meaningfully compare welfare across counterfactuals because these interventions are implemented conditional on the existing information environment—that is, incorporating students’ biases. As such, any differences relative to the baseline reflect changes in the mechanism or reapplication rules rather than differences in the information available to students.

*Assignment Mechanisms.* We evaluate the effects of eliciting the intensity of students’ preferences through changes in the assignment mechanism using two alternative approaches (see Appendix F.1):

1. Constrained Deferred Acceptance (CDA): Change the constraint in the length of the ROLs,  $K$ . We evaluate  $K \in \{1, 2, 3\}$ , since most students submit an ROL with length less than or equal to 3.
2. Choice-Augmented Deferred Acceptance (CADA): Students can signal one program on their ROL and receive a score bonus (score multiplier  $\psi > 1$ ) applied to their high school GPA component. We implement this mechanism only for first-period applicants; consequently, students who apply in the second period do not receive the bonus.

Both mechanisms elicit the intensity of students’ preferences, since they introduce opportunity costs that students must take into account when submitting their applications. In the case of CDA, constraining the length of applicants’ lists limits students in including other programs on their ROLs, so they must account for the opportunity cost of including each program. In the case of CADA, students can signal only one program, so they must carefully decide which program to target to get the bonus.

Notice that eliciting the intensity of students’ preferences may not necessarily lead to higher yield. On the one hand, if eliciting this information decreases initial mismatches, we would expect to reduce inefficient switching. On the other hand, if the assignment mechanism also elicits the intensity of preferences of students who reapply to the system and

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<sup>9</sup>Ex-post utilities are computed at the end of period two, adding the discounted value function of period three—i.e., after students have made all of their choices in the model.

these change considerably due to learning, we would see an increase in efficient switching due to a higher value of reapplication. In this sense, we expect that under CADA—which provides a score bonus only in the first period—switching would decrease more than in the case in which the score bonus is applied in both periods, because the policy also gives a comparative advantage to first-period applicants, which increases switching costs through the higher equilibrium cutoff scores produced by the bonus.

*Reapplication Rules.* Another policy to reduce the incentives to switch is to provide bonuses to students applying for the first time or to penalize students who reapply and try to switch programs; these policies have been implemented in Finland and Turkey, respectively. We implement these policies in our counterfactual analysis as follows, where  $\psi$  represents a score multiplier applied to the high school GPA component:

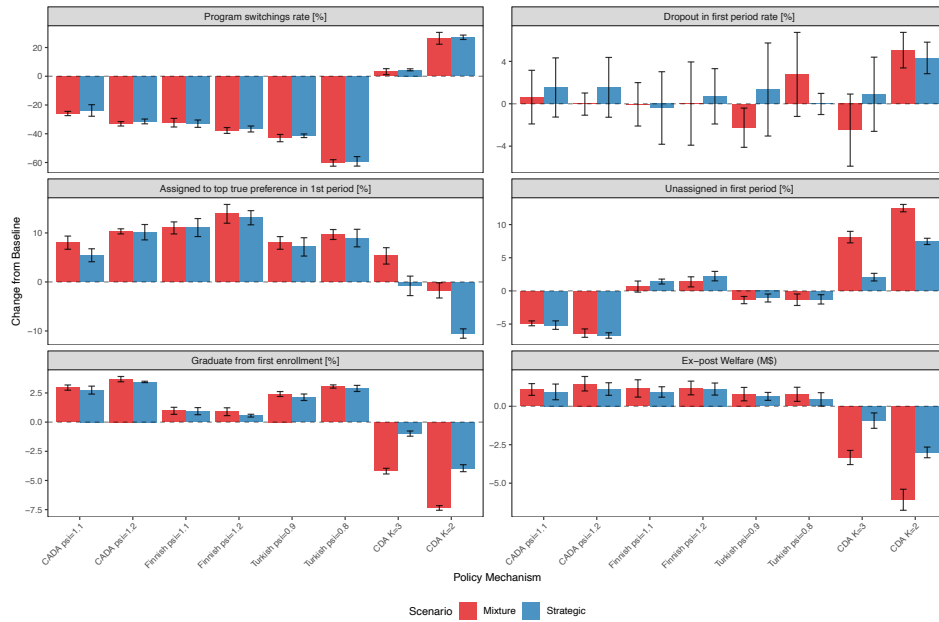
- (i) Turkish rule: Applicants currently enrolled in the centralized system receive a score penalty (score multiplier  $\psi < 1$ ).
- (ii) Finnish rule: First-time applicants receive a score bonus (score multiplier  $\psi > 1$ ).

Even though both policies aim to reduce the incentives for switching, they affect students' applications and reapplications in different ways. On the one hand, the Finnish policy directly reduces the incentives to reapply to the system, regardless of the programs students include in their ROLs. As a result, the Finnish policy increases the continuation value of choosing the outside option, and thus increases the fraction of students who wait an extra year to submit their first application. On the other hand, the Turkish policy reduces the incentives to reapply if students were previously enrolled in a program in the system—i.e., it reduces the incentives to apply to programs if they are very likely to switch from them in the future (e.g., programs for which students have low preference intensity). Hence, the Turkish policy may decrease the fraction of students who enroll in the first period in less preferred programs. Despite these differences, we expect that both policies would decrease the frequency of reapplications and switches. In contrast, the welfare effects of these policies is unclear. Students may benefit from these policies, since both the penalty and the bonus help to address the negative externality that switchers generate in the system. How-

ever, since under these policies students face more barriers to switching, the benefits of learning become lower, so students' welfare may decrease.

*Results.* In Figure 5, we present a summary of the counterfactual results as percentage changes relative to the baseline. The detailed results are reported in Tables F.III and F.IV in Appendix F. Since students' application behavior—i.e., whether they act strategically or as weak truth-tellers—might not be invariant to counterfactuals that heighten strategic incentives, we present bounds on policy effects by considering two benchmark scenarios: (i) all students acting strategically ( $\rho = 0$ ), and (ii) students behaving according to a mixture with probability of strategic behavior given by that estimated in the baseline model ( $\rho = \hat{\rho}$ ).

FIGURE 5.—Summary of Counterfactual Results



On the one hand, we find that implementing CDA increases the share of students who remain unassigned, switch programs, and drop out in the first period. This pattern is intuitive: reducing the maximum length of students' ROLs raises the risk of remaining unassigned and generates more initial mismatches, thereby increasing the incentives to switch or drop

out. While the program switching rate (conditional on assignment) increases monotonically as  $K$  decreases, the absolute number of program switchers peaks at  $K = 2$  and then declines at  $K = 1$  due to a sharp rise in unassigned students, which shrinks the pool of potential switchers. Moreover, CDA significantly decreases the share of students assigned to their top true preference—particularly when students behave strategically, as they may skip preferred programs to improve their chances of admission elsewhere. Consequently, CDA leads to significantly lower graduation rates and lower ex-post welfare. Consistent with this, ex-post welfare declines as  $K$  decreases.

On the other hand, CADA and the policies involving changes to the re-application rules can improve student outcomes. From the top panel—showing the percentage change in switching and dropout rates relative to the baseline—we observe that all these policies lead to a sizable reduction in program switching, with the largest effect under the Turkish policy.

From the middle panel—reporting the percentage change in the probability of assignment to the top true preference and the probability of being unassigned in the first period relative to the baseline—we find that CADA and the re-application rule changes significantly increase the share of students assigned to their top true preference, with the Finnish and CADA policies delivering comparable improvements. In the fully strategic case, we also observe that changes to the re-application rules cause a small increase in the share of unassigned students in the first period, whereas CADA produces a slight decrease.

From the bottom panel—showing the percentage change in the probability of graduating from the first enrollment and the change in ex-post welfare relative to the baseline—we observe that these policies lead to a significant increase in the probability of graduation from the first enrollment, with the largest gains under CADA and the Turkish policy. When considering the mixture of student types, ex-post welfare increases under all these policies, with CADA generating the highest welfare improvement, and the Finnish policy being the best at eliciting preference intensity—assigning students to their top-true preference.

Finally, regarding student types, Tables F.III and F.IV in Appendix F reveal that males generally benefit more than females across the different policies. However, the effects



across income groups depend on the policy's intensity, underscoring that policy design may entail trade-offs in equity and distributional fairness.

Overall, these results indicate that restricting the length of applicants' lists can have detrimental effects on both switching behavior and student welfare. In contrast, CADA and the re-application rule policies appear promising for increasing the system's yield, mitigating the congestion externalities caused by initial mismatches, and enhancing welfare. However, the optimal policy depends on the policymaker's objective: if the goal is to reduce switching and increase system yield, the Turkish policy performs best, whereas if the aim is to maximize student welfare, CADA delivers superior outcomes.

## 8. CONCLUSIONS

In this paper, we analyze the determinants of students' college progression in the presence of a centralized assignment mechanism. We aim to understand why students reapply, switch, or drop out, and how the incentives generated by the mechanism and the learning environment shape these outcomes and students' welfare. We therefore study admissions as a dynamic environment with information frictions, belief updating about abilities and preferences, strategic incentives, and continued interaction with the mechanism over time.

Using data from the Chilean college admissions system and multiple nationwide surveys that we designed and conducted, we provide empirical evidence showing that two central behavioral channels explain students' dynamic decisions. The first channel, the *initial mismatch* channel, predicts that students may have incentives to switch programs if they were initially assigned to less desired preferences. The second channel, the *learning* channel, encompasses both learning about abilities and preferences over majors, leading students to switch to programs where expected outcomes—and their preference intensities—are higher.

Given these findings, we introduce a structural model that captures students' decisions during their academic progression, allowing them to learn about their abilities from grades and about their preferences from initial enrollment, while holding heterogeneous and potentially biased beliefs over college outcomes and labor-market returns. We use the estimated model to evaluate (i) changes to the learning environment—such as counseling in

high school and a mandatory year of a post-secondary program—and (ii) different market design choices—including changes in the assignment mechanism such as adding further constraints on list length and allowing a single signal that grants a score bonus, or modifying the reapplication rules to match those used in Turkey and Finland. Our results show that the former can increase the system’s yield, while reapplication rules and the signaling mechanism can also increase students’ welfare. Importantly, these effects are robust to changes in the fraction of participants who behave strategically, unlike other approaches such as constraining list lengths. The best policy ultimately depends on the planner’s objective and the trade-off between yield and the distribution of welfare gains across students.

Overall, our results show that incorporating dynamic incentives and eliciting information on participants’ cardinal preferences can significantly increase students’ welfare and improve downstream outcomes. These insights can be informative for improving the design of many matching markets that exhibit similar features. For instance, in organ transplant systems, patients have private information regarding their health, face dynamic considerations such as when to accept or reject an organ, and even learn about organs’ qualities over time (Zhang, 2010), all of which may affect patient survival (Agarwal et al., 2021). In entry-level labor markets, candidates have private information about their preferences, learn about their match qualities through experience, and face dynamic considerations, such as deciding when to enter the market (apply), re-enter (reapply), exit (dropout), or rematch (switch), which affect employers’ retention in turn. Our key insight is that market designers should correctly balance the gains from learning through experience and the crowd-out externality produced by initial mismatches to improve the efficiency of these markets.

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